

Syllabus

Fall semester 2025

Course Title	Chemistry and Experiment II	Course Code	CB1111		Credit	3	Classroom	Tamgu-gwan 2106
		Type	Core Curriculum		Hours	3		
Teacher	Name	Class no.	Office	Office hours	Contact		E-mail	
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1. Course Description

The Chemistry and Experiment II course aims to deliver comprehensive understanding in chemical change and reactions. The macro- and microscopic properties of liquids, solids, and solution will be explored. The kinetics, chemical equilibria, and spontaneity of various reaction systems will be addressed subsequently. The course will outline the introduction of electrochemistry regarding oxidation and reduction process, and how they are related with thermochemistry. It is anticipated that the students will gain (a) fundamental concepts in chemical transformation, and (2) mathematical skills to interpret various situations.

2. Core Target Competencies

Subject Core Competencies						
Scientific Inquiry & Research	Mathematical & Computational Reasoning	Graphical & Explanatory Reasoning	Critical Thinking & Problem Solving	Communication & Presentation	Global & Collaborative Leadership	Personal Development & Well-being
1. Scientific Inquiry & Research: Focuses on the skills necessary for rigorous scientific investigation, from formulating questions to disseminating findings. 2. Mathematical & Computational Reasoning: Emphasizes applying advanced mathematical principles and computational thinking to solve complex problems. 3. Graphical & Explanatory Reasoning: Emphasizes constructing graphical models and detailed written explanations during problem solving. 4. Critical Thinking & Problem Solving: This domain highlights the ability to deeply analyze information, evaluate arguments, and generate effective solutions. 5. Communication & Presentation: Covers effective expression and understanding across various modalities and audiences, both written and oral. 6. Global & Collaborative Leadership: This domain encompasses the capacity to lead, collaborate effectively with diverse groups, and engage responsibly with global and community challenges to foster positive change. 7. Personal Development & Well-being: Encompasses self-management, resilience, and a commitment to holistic physical, mental, and emotional health.						

3. Textbooks and References

A. Textbook: Chemistry; Principles and Reactions, 8th Edition – by Masterton and Hurley (Cengage Learning)

B. References

- Principle of Modern Chemistry, 8th Edition – by Oxtoby, Gillis, and Butler (Cengage Learning)

4. Class Method

- (1) The diversity of the class members ought to be respected. Any type of discrimination regarding ethnicity, nationality, language, personal orientation, etc., will not be tolerated.
- (2) The lectures and all related communications will be conducted in English as part of addressing the language diversity of all members. Other languages are not allowed unless permission is granted by the instructor.
- (3) The lectures will be delivered in class by instructor's presentation. Students are highly recommended to take handwritten notes in every class to enhance their summarization, clarification, and graphical description skills.
- (4) A chapter summary will be required from the students prior to the beginning of each chapter.
- (5) A problem set will be given to the students after finishing each chapter in class. The students are required to solve and submit the given problems to enhance their complex problem-solving skills.
- (6) The students are required to participate in discussions regarding the given topics and questions to enhance their communication and critical thinking skills.
- (7) Assessment will be performed based on both examination and non-examination criteria. Two exams (mid-term and final) will be conducted as part of the examination evaluation. Laboratory activities, assignments, attendance, attitude, and in-class participation will be evaluated as part of the non-examination criteria.
- (8) All of the students' evaluation materials (e.g., lecture notes, problem sets, project outcomes, etc.) must be submitted to the Google Classroom.
- (9) Any late submissions after the due date or time will not be accepted, resulting in 0 point for the related criteria.
- (10) Any type of misconduct (e.g., cheating, copying, refusing to follow instructions, dishonorable actions, other unauthorized activities, etc.) will automatically result in an F grade for the course. Furthermore, the student will receive official school punishment according to the Korea Science Academy Regulations.

5. Evaluation

A. Evaluation Criteria

Examination (60)		Non-examination (40)			
Mid-term Exam	Final Exam	Laboratory Activities	Summary and Problem Set Assignments	Attitude and In-class Participation	Attendance
30	30	15	15	5	5

B. Evaluation Method

- (1) Mid-term and Final Exam (30 points each; total 60 points for the Examination criteria)

One mid-term and one final exam will be held after covering specific chapters. Understanding basic knowledge, creative thinking, cognitive flexibility, literal explanation, and graphical demonstration will be important competencies for the evaluation.

Each exam will be held for approximately 180 minutes. The exam duration may be adjusted based on the given circumstances.

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unauthorized activities, etc.) will automatically result in an F grade for the course. Furthermore, the student will receive official school punishment according to the Korea Science Academy Regulations.

(2) Laboratory Activities (15 points)

Each team of two to three students will perform laboratory activities regarding the specific topics.

The laboratory reports will be evaluated in three categories; (a) pre-reports for the preparation, (b) in-lab reports written during the experiment, and (c) final reports to summarize and discuss the overall laboratory activities.

The students are required to follow the lab safety rule. Disobeying the safety instructions will be penalized which may result in 0 point for the related criteria.

The reports will be summarized in handwritten format and submitted as a PDF file.

The content of submitted PDF files must be clear and easy to read. Any vague content in terms of light and dark contrast, color, etc., will not be assessed, resulting in 0 point for the related criteria. The use of a scanner is recommended over the use of a cell phone camera.

Any late submissions after the due date or time will not be accepted, resulting in 0 point for the related criteria.

Any type of misconduct (e.g., cheating, copying, refusing to follow instructions, dishonorable actions, other unauthorized activities, etc.) will automatically result in an F grade for the course. Furthermore, the student will receive official school punishment according to the Korea Science Academy Regulations.

(3) Summary and Problem Set Assignments (15 points)

Students are required to submit handwritten summary prior to the beginning of each chapter. Each summary must be submitted as a single PDF file. Taking handwriting on tablets (e.g., iPad, Galaxy Tab, etc.) is allowed. However, lecture notes created using word processing software (e.g., Microsoft Word, Pages, Hangeul, etc.) are not permitted. The graphical figures must be drawn by hand. This policy is intended to enhance students' summarization and graphical description skills.

Students will receive problem set assignments after finishing each chapter. The problem sets must be solved and submitted to the instructor as a single PDF file.

The content of submitted PDF files must be clear and easy to read. Any vague content in terms of light and dark contrast, color, etc., will not be assessed, resulting in 0 point for the related criteria. The use of a scanner is recommended over the use of a cell phone camera.

Any late submissions after the due date or time will not be accepted, resulting in 0 point for the related criteria.

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(4) Attitude and In-class Participation (5 points)

Students will start with 0 points in the Attitude and In-class Participation criteria. Students may earn points through positive behaviors and actions, while negative actions and behaviors will result in point deductions.

Attitude will be evaluated as follows: Positive contributions to the learning environment may be rewarded with +0.5 points. Behaviors that disrupt the learning environment will be penalized -0.5 points per occurrence (e.g., unauthorized use of a non-English language, studying for other classes, unauthorized chatting, taking a nap, etc.). Unauthorized use of cell phones and computers will result in a -2 point deduction per occurrence. Unauthorized consumption of food and beverages is not permitted in the classroom.

In-class participation will be evaluated as follows: Active participation and communication during class may result in earning positive points. Low participation and weak involvement in discussion will lead to the point deductions.

The assessment of the Attitude and In-class Participation criteria will be subject to the instructor's discretion.

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(5) Attendance and Attitude (5 points)

Attendance will be recorded and graded as follow: Students will receive 5 points for no unexcused absence and no late arrival. Being late will be penalized by -0.5 point per each class. One unexcused absence will be penalized by 1 point. Two unexcused absences will be penalized by 3 points. Three unexcused absences will be penalized by 5 points and will not receive any points in Attendance criterion. Four or more unexcused absences will be result in F grade of the course.

Any type of misconduct (e.g., cheating, copying, refusing to follow instructions, dishonorable actions, other unauthorized activities, etc.) will automatically result in an F grade for the course. Furthermore, the student will receive official school punishment according to the Korea Science Academy Regulations.

6. Criteria for Awarding Credits

- ☞ The grade will be given as PASS/FAIL according to the received total points. The overall points equal to or higher than 70 will receive PASS grade; The total points lower than 70 will receive FAIL grade.
- ☞ **Students with excellent academic achievement are recommended for the Academic Excellence Award and the Principal's Award, and are recorded according to core competencies in the Subject Details. However, students with poor academic attitude or poor attendance are excluded.**
- ☞ **Any type of misconduct (e.g., cheating, copying, refusing to follow instructions, dishonorable actions, other unauthorized activities, etc.) will automatically result in an F grade for the course. Furthermore, the student will receive official school punishment according to the Korea Science Academy Regulations.**

- Attendance for at least three-quarters of the total class hours is required to earn credit.
- For Pass/Fail courses, if the instructor determines that a specific evaluation element is insufficient and requires a separate evaluation, a separate evaluation may be conducted.
- Calculation of the basis score for awarding credit to absentees : For examinations missed due to an officially excused absence or a medically certified absence, the recognized score shall be determined as follows:
(the percentile score corresponding to the missed examination, calculated based on the total scores obtained by the student in other examinations of the same subject) × (100% for an excused absence; 80% for a medical absence).
- If a basis score cannot be calculated, the grading method : (Not applicable)
- ※ Refer to the "Academic Performance Management Regulations" for the percentage of credit points applied.

7. Weekly Plans

Week (period)	Topic and Contents	(Notes)
Week 1 8.18.~8.22.	Chapter 9. Liquids and Solids	
Core Competencies	♦ Scientific Inquiry & Research, Mathematical & Computational Reasoning, Graphical & Explanatory Reasoning, Critical Thinking & Problem Solving, Communication & Presentation, Global & Collaborative Leadership, Personal Development & Well-being:	
Week 2 8.25.~8.29.	Chapter 10. Solutions Laboratory Activity 1. Lab Safety	
Core Competencies	♦ Scientific Inquiry & Research, Mathematical & Computational Reasoning, Graphical & Explanatory Reasoning, Critical Thinking & Problem Solving, Communication & Presentation, Global & Collaborative Leadership, Personal Development & Well-being:	
Week 3 9.1.~9.5.	Chapter 11. Rate of Reaction	
Core Competencies	♦ Scientific Inquiry & Research, Mathematical & Computational Reasoning, Graphical & Explanatory Reasoning, Critical Thinking & Problem Solving, Communication & Presentation, Global & Collaborative Leadership, Personal Development & Well-being:	
Week 4 9.8.~9.12.	Chapter 11. Rate of Reaction Chapter 12. Gaseous Chemical Equilibrium	
Core Competencies	♦ Scientific Inquiry & Research, Mathematical & Computational Reasoning, Graphical & Explanatory Reasoning, Critical Thinking & Problem Solving, Communication & Presentation, Global & Collaborative Leadership, Personal Development & Well-being:	
Week 5 9.15.~9.19.	Chapter 12. Gaseous Chemical Equilibrium	
Core Competencies	♦ Scientific Inquiry & Research, Mathematical & Computational Reasoning, Graphical & Explanatory Reasoning, Critical Thinking & Problem Solving, Communication & Presentation, Global & Collaborative Leadership, Personal Development & Well-being:	
Week 6 9.22.~9.26.	Chapter 12. Gaseous Chemical Equilibrium Chapter 13. Acids and Bases	

Core Competencies	♦ Scientific Inquiry & Research, Mathematical & Computational Reasoning, Graphical & Explanatory Reasoning, Critical Thinking & Problem Solving, Communication & Presentation, Global & Collaborative Leadership, Personal Development & Well-being:	
Week 7 9.29.~10.3.	Chapter 13. Acids and Bases	National Foundation Day (10.03.)
Core Competencies	♦ Scientific Inquiry & Research, Mathematical & Computational Reasoning, Graphical & Explanatory Reasoning, Critical Thinking & Problem Solving, Communication & Presentation, Global & Collaborative Leadership, Personal Development & Well-being:	
Week 8 10.6.~10.10.	Chuseok Holiday, Hangeul Day, Discretionary Holiday	

Week 9 10.13.~10.17.	Midterm Exams (10.13.~10.16.)	10.17. Regular classes
Week 10 10.20.~10.24.	Chapter 14. Equilibria in Acid-Base Solutions	
Core Competencies	♦ Scientific Inquiry & Research, Mathematical & Computational Reasoning, Graphical & Explanatory Reasoning, Critical Thinking & Problem Solving, Communication & Presentation, Global & Collaborative Leadership, Personal Development & Well-being:	
Week 11 10.27.~10.31.	Chapter 14. Equilibria in Acid-Base Solutions Laboratory Activity 2. Acid-base Titrations	
Core Competencies	♦ Scientific Inquiry & Research, Mathematical & Computational Reasoning, Graphical & Explanatory Reasoning, Critical Thinking & Problem Solving, Communication & Presentation, Global & Collaborative Leadership, Personal Development & Well-being:	
Week 12 11.3.~11.7.	Chapter 16. Spontaneity of Reaction	
Core Competencies	♦ Scientific Inquiry & Research, Mathematical & Computational Reasoning, Graphical & Explanatory Reasoning, Critical Thinking & Problem Solving, Communication & Presentation, Global & Collaborative Leadership, Personal Development & Well-being:	
Week 13 11.10.~11.14.	Chapter 16. Spontaneity of Reaction	

Core Competencies	♦ Scientific Inquiry & Research, Mathematical & Computational Reasoning, Graphical & Explanatory Reasoning, Critical Thinking & Problem Solving, Communication & Presentation, Global & Collaborative Leadership, Personal Development & Well-being:	
Week 14 11.17.~11.21.	Chapter 17. Electrochemistry	
Core Competencies	♦ Scientific Inquiry & Research, Mathematical & Computational Reasoning, Graphical & Explanatory Reasoning, Critical Thinking & Problem Solving, Communication & Presentation, Global & Collaborative Leadership, Personal Development & Well-being:	
Week 15 11.24.~11.28.	Chapter 17. Electrochemistry	
Core Competencies	♦ Scientific Inquiry & Research, Mathematical & Computational Reasoning, Graphical & Explanatory Reasoning, Critical Thinking & Problem Solving, Communication & Presentation, Global & Collaborative Leadership, Personal Development & Well-being:	
Week 16 12.1.~12.5.	Laboratory Activity 3. Electrolysis of Water	Make-up class week
Week 17 12.8.~12.12.	Final Exams (12.08.~12.12.)	

※ The above plans may change according to the academic schedule.

[Reference] Syllabus Competence Reference

1. Analytical and Creative Thinking	2. Complex Communication	3. Leadership and Teamwork	4. Digital/ Quantitative Literacy	5. Global Perspective	6. Adaptability/ Progression/ Spirit of Adventure	7. Integrity and Ethical Decision Making	8 Habits/ Mindset
Ability to identify, manage and deal with complex problems	Ability to understand and express themselves in two or more languages	Ability to be the first to come up with and present new ideas	Ability to understand, use and apply digital technology	Ability to develop an open mind about the traditions or values of others	Ability to develop flexibility, agility and adaptability	Ability to sustain an empathetic and understanding perspective	Sincerity
Ability to see through prejudice and identify reliable information and uncertain information	Ability to communicate clearly with diverse audiences or readers	Ability to influence people to follow themselves, rather than lead them through directives, orders, or authority	Ability to create digital contents and media	Ability to understand the history, politics, religion, and culture of another region	Ability to challenge unfamiliar situations	Ability to cultivate sincerity, honesty, fairness, and respect creativity	Creativity
Ability to find quality information in the flood of information	Ability to listen attentively	Ability to build trust, resolve conflicts, and support others	Ability to utilize multimedia resources to effectively communicate one's ideas in a variety of formats	Ability to use one or more of the international languages used worldwide	Ability to explore and experiment	Ability to display moral courage in the face of injustice	A love of learning and curiosity
Ability to articulate meaningful questions	Ability to speak effectively and coherently	Ability to facilitate group discussions or discussions, build consensus, and negotiate outcomes	Ability to master and use higher mathematics	Ability to use technology to connect people and things around the world	Ability to word effectively in situations of ambiguous or changing priorities	Ability to act responsibly for the benefit and happiness of the greater community	Resilience/elasticity
Ability to analyze and create ideas or knowledge	Ability to write concisely and clearly, considering the diversity of readers	Ability to teach, guide, and counsel Ability to seek help from others	Ability to understand traditional or recently emerging fields such as mathematics, science, technology, environmental science, robotics, geometry, cellular automata, nanotechnology, and biotechnology	Social and intellectual skills to effectively navigate information across multiple cultures	Ability to see failure as a lesson and to accept that innovation is made up of frequent failures and small successes	Ability to have an intrinsic understanding of ethical issues or dilemmas about new media or new technologies	Perseverance/patience
Ability to devise and test solutions to problems by trial and error	Ability to explain information and persuasively engage others in an engaging way	Ability to collaborate on tasks, manage groups, and delegate responsibilities		Ability to use 21 st century core competencies such as communication skills, collaboration skills, critical thinking skills, creativity to properly understand and deal with global issues	Ability to develop a sense of independence to seek new roles, ideas, or strategies	Ability to make rational and ethical decisions on complex issues	Self-efficacy

1. Analytical and Creative Thinking	2. Complex Communication	3. Leadership and Teamwork	4. Digital/ Quantitative Literacy	5. Global Perspective	6. Adaptability/ Progression/ Spirit of Adventure	7. Integrity and Ethical Decision Making	8 Habits/ Mindset
Ability to think of different ways to cope		Ability to listen and practice goals		Ability to learn from and collaborate with people from different cultures, religions and lifestyle through mutual respect and open dialogue	Ability to develop the knowledge and competencies necessary to start a business and run a business		Stress management skills
Ability to develop knowledge and perspectives that are not limited to one field but span multiple fields		Ability to share credit for success		Ability to generate diverse ideas and exploit social and cultural differences to get things done			Time management skills
Ability to make sustained inferences							
Ability to put things together or adapt them to new uses and situations							
Ability to solve new problems that cannot be solved through established rules							
Ability to use knowledge or creativity to solve complex real-world problems							

1. Complex Problem-Solving Skills	Ability to solve new and unclear problems that may arise in complex real-world situations
2. Critical Thinking Skills	Ability to compare and analyze the strengths and weaknesses of various solutions or approaches to a given problem using logic and reasoning
3. Creativity	Ability to come up with novel ideas when given a certain topic or situation or to suggest a creative method in the process of solving a problem
4. Human Resource Management Skills	Ability to select the most suitable person for a specific task and manage them to achieve maximum work performance through motivation, and systematic efficient management
5. Ability to Collaborate	Ability to adjust one's own behavior in response to the behavior of others
6. Emotional Ability	Ability to perceive other people's reactions and understand why they react the way they do
7. Judgement and	Ability to make the most rational judgement and decision by analyzing the costs and benefits

Decision-making Skills	of a specific situation or problem
8. Orientated Towards Service	Ability to actively seek ways to help others
9. Negotiation Skills	Ability to bring together parties who differ in ideas or interests and try to reduce those differences
10. Cognitive Flexibility	Ability to apply different kinds of rules or principles to rearrange and reorganize things in different ways